AG News Classification

Exploratory Data Analysis Report

Preparing Data for Traditional ML, Deep Learning, and Transformer-Based Models

2026-05-04

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1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the AG News Classification Dataset, which contains 89,320 news articles categorized into four classes: World (Label 1), Sports (Label 2), Business (Label 3), and Sci/Tech (Label 4). The primary objective of this analysis is to provide actionable insights for data cleaning, preprocessing, and model architecting across three distinct modeling paradigms: Traditional Machine Learning (TF-IDF / CountVectorizer with classifiers like Logistic Regression, SVM, and Random Forest), Deep Learning (LSTM, GRU, CNN for text), and Transformer-based Fine-tuning (BERT).

The dataset exhibits a perfectly balanced class distribution with 22,330 samples per class, which eliminates the need for oversampling, undersampling, or class-weight adjustments during training. However, the analysis reveals several critical data quality issues that must be addressed before modeling. Approximately 29.6% of samples contain HTML entities (e.g., #39;, <, >), 3.1% contain URLs or web links, and 10.7% include source attribution tags such as "Reuters" or "AP" that could serve as spurious correlation features. These artifacts, if left untreated, will degrade model generalization and introduce bias toward source-based patterns rather than content-based classification.

Text length analysis shows that articles average 37.9 words (median 37) with a standard deviation of 10.1, and the 95th percentile falls at approximately 56 words. This distribution is well-suited for BERT with a max_seq_len of 128, which would capture 99%+ of all samples without truncation. For deep learning models, a padding length of 64 would cover approximately 90% of samples, while 80-100 tokens would be optimal for full coverage. The vocabulary comprises approximately 135,689 unique tokens across the corpus, with the Sci/Tech class exhibiting the highest lexical diversity (59,191 unique tokens) and Business the lowest (48,179 unique tokens).

Key Findings at a Glance

- **Balanced Dataset:** Perfectly balanced with 22,330 samples per class (25% each) — no resampling needed.
- **HTML Contamination:** 29.6% of samples contain HTML entities (#39;, <, >) requiring cleanup.
- **Source Tag Leakage Risk:** 10.7% of samples contain "Reuters", 7.1% contain "AP" — potential data leakage features that should be stripped.
- **URL Artifacts:** 3.1% of samples contain URL fragments (http, www, .com) — especially prevalent in Business class.
- **Optimal Sequence Length:** BERT max_seq_len=128 captures 99%+ of samples; LSTM padding=80-100 tokens recommended.
- **Vocabulary Size:** ~135K unique tokens; top-10K vocab sufficient for Traditional ML; Sci/Tech has highest lexical diversity.
- **Discriminative Features:** Class-specific bigrams detected (e.g., "prime minister" for World, "red sox" for Sports, "oil prices" for Business, "microsoft" for Sci/Tech).

2. Dataset Overview & Definitions

The AG News Classification Dataset is derived from the AG News corpus, which was originally constructed by assembling news articles from over 2,000 news sources. The version analyzed in this report (train_split.csv) contains 89,320 samples, each consisting of a text field with the concatenated news title and description, along with an integer label in the range [1, 4]. This split represents a training partition of the dataset, commonly used for model development and validation through cross-validation or hold-out strategies.

Field	Type	Description	Example
text	string	Concatenated title + description	A stinker? Critics pan Spacey #39;s Old Vic debut...
label	int64	Class label (1=World, 2=Sports, 3=Business, 4=Sci/Tech)	4

Table 1: Dataset Schema

Label	Class Name	Sample Count	Proportion	Description
1	World	22,330	25.0%	International news, politics, conflicts, diplomacy
2	Sports	22,330	25.0%	Athletics, games, tournaments, player updates
3	Business	22,330	25.0%	Finance, markets, corporate news, economy
4	Sci/Tech	22,330	25.0%	Technology, science, software, innovation

Table 2: Label Mapping and Class Descriptions

The dataset was originally extracted from news aggregation services and therefore contains artifacts typical of web-scraped data, including HTML entities, source attribution tags (AP, Reuters, AFP), escaped characters, and occasional URL fragments. Understanding these characteristics is essential for designing an effective preprocessing pipeline that preserves discriminative content while removing noise. The four classes represent high-level topical categories that are roughly comparable in scope, which explains the balanced distribution observed in the curated version of the dataset.

3. Data Quality Assessment

A rigorous data quality assessment forms the foundation of any reliable machine learning pipeline. This section evaluates the dataset across multiple quality dimensions including completeness, uniqueness, consistency, and the presence of noise artifacts. Understanding these quality metrics is critical for determining the extent of preprocessing required and for anticipating potential failure modes during model training and inference.

3.1 Completeness & Uniqueness

Quality Metric	Value	Assessment
Total Samples	89,320	Large dataset — sufficient for all 3 modeling paradigms
Missing Values (text)	0 (0.0%)	Complete — no imputation needed
Missing Values (label)	0 (0.0%)	Complete — no label imputation needed
Exact Duplicate Rows	0 (0.0%)	No duplicates — no deduplication required

Quality Metric	Value	Assessment
Duplicate Texts	0 (0.0%)	No text collisions — each article is unique
Empty/Whitespace Texts	0 (0.0%)	No empty strings detected

Table 3: Completeness and Uniqueness Assessment

The dataset achieves perfect scores on completeness and uniqueness metrics. There are zero missing values in both the text and label columns, zero exact duplicate rows, and zero duplicate texts. This indicates that the dataset has been well-curated at the sampling level. However, as we will see in subsequent sections, the absence of missing values does not imply the absence of noise — the text content itself contains numerous artifacts that require attention during preprocessing.

3.2 Text Content Anomalies

Anomaly Type	Affected Rows	Prevalence	Risk Level
HTML Entities (#39;, etc.)	26,405	29.6%	HIGH
Source Tags (Reuters, AP, AFP)	16,589	18.6%	HIGH
URL/Link Fragments	2,787	3.1%	MEDIUM
Escaped Dollar Sign	218	0.2%	LOW
#NAME? Artifacts	20	0.02%	LOW
Special Characters	89,193	99.9%	MEDIUM
Numeric Values	51,881	58.1%	LOW-MEDIUM

Table 4: Text Content Anomaly Summary

The most prevalent data quality issue is the presence of HTML entities, affecting nearly 30% of all samples. The most common entity is #39; (apostrophe encoding), which appears 33,036 times across the corpus. This is followed by < and > (HTML tag brackets), which appear 9,802 times each and are remnants of HTML markup that was not properly stripped during data collection. These entities create tokenization noise and inflate vocabulary size, making them a high-priority target for preprocessing.

Source attribution tags represent a more insidious data quality issue. While they do not introduce noise per se, they create a risk of spurious correlation: models may learn to associate "Reuters" with Business (where it appears in 18.4% of samples) or "AP" with Sports (10.7%), rather than learning the actual content patterns. This is a form of data leakage that can inflate training accuracy while degrading real-world generalization. The recommended approach is to remove these source tags during preprocessing, or at minimum, to evaluate model robustness by testing on samples with and without source attribution.

4. Class Distribution Analysis

Class distribution analysis is one of the most critical steps in any classification task, as it directly impacts model training dynamics, loss function design, and evaluation metric selection. An imbalanced dataset may require techniques such as class weighting, oversampling (SMOTE), or stratified sampling to prevent the model from being biased toward majority classes. Fortunately, the AG News dataset exhibits a perfectly balanced distribution across all four classes, with exactly 22,330 samples per category.

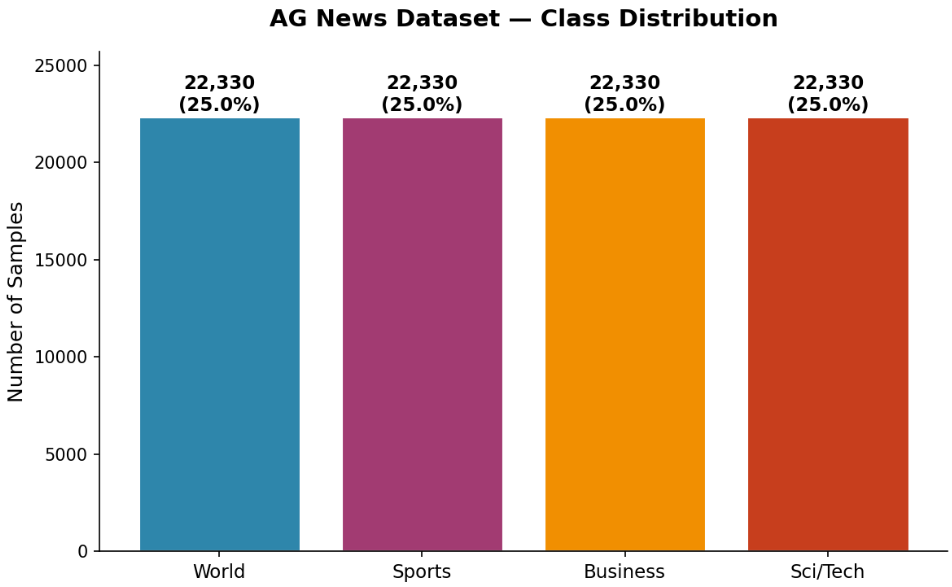

Figure 1: Class Distribution — Perfectly Balanced with 22,330 Samples per Class

The perfectly balanced distribution means that standard training procedures can be applied without any class imbalance mitigation strategies. Accuracy is a reliable primary metric, though it should be supplemented with per-class precision, recall, and F1-score to detect potential class-specific weaknesses. No oversampling, undersampling, or class-weight adjustments are needed for any of the three modeling approaches. This balance also ensures that stratified k-fold cross-validation will produce reliable estimates of model performance.

5. Text Length & Statistical Analysis

Understanding text length distributions is essential for configuring model architectures and preprocessing parameters. For Traditional ML approaches, text length affects TF-IDF sparsity and n-gram coverage. For Deep Learning models (LSTM, CNN), it determines padding/truncation thresholds and batch memory requirements. For Transformer models (BERT), it directly constrains the max_seq_length parameter, which has significant implications for both computational cost and information retention.

5.1 Overall Text Length Statistics

Statistic	Character Length	Word Count
Mean	236.5	37.9
Median	232.0	37.0
Std Dev	66.5	10.1
Min	17	4

Statistic	Character Length	Word Count
25th Percentile	196	32
75th Percentile	266	43
95th Percentile	351	56
99th Percentile	469	69
Max	1,012	177

Table 5: Overall Text Length Statistics

The average article contains 37.9 words (approximately 236 characters), with a standard deviation of 10.1 words. The distribution is slightly right-skewed, with a few outlier articles extending to 177 words. The interquartile range (IQR) spans 11 words (32-43), indicating that the majority of articles are relatively compact. This is consistent with the dataset being composed of news title + description pairs rather than full-length articles. The compact nature of these texts makes them particularly well-suited for Transformer models, which can process most samples within the standard 128-token BERT limit without any truncation.

5.2 Word Count Distribution by Class

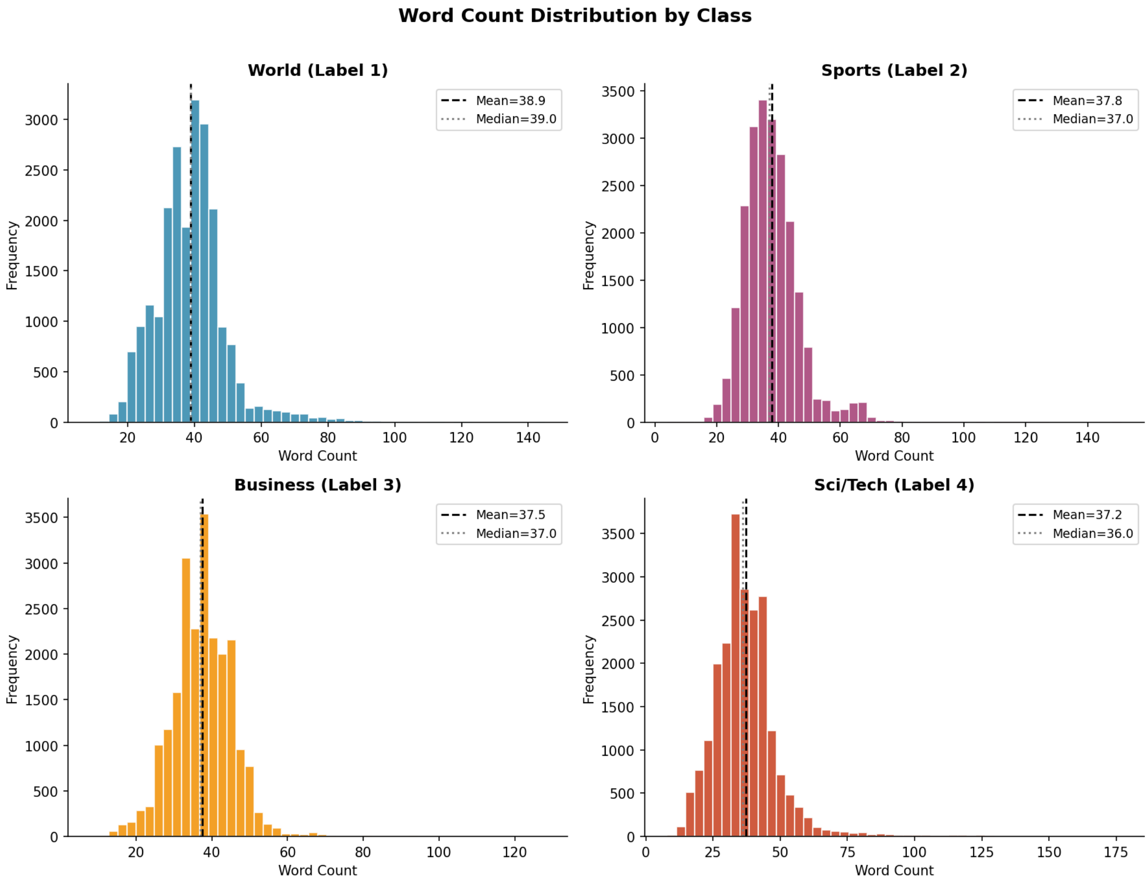

Figure 2: Word Count Distribution by Class — Histograms with Mean and Median

The per-class word count distributions reveal subtle but meaningful differences across categories. World news articles tend to be slightly longer on average (38.9 words) compared to Sci/Tech articles (37.2 words), though the difference is modest (Cohen's d approximately 0.17, negligible effect size). All four distributions exhibit a similar right-skewed shape with the bulk of the mass concentrated between 25-55 words. The Sports class shows a slightly wider spread, possibly reflecting the variety of content types (game recaps, player profiles, tournament previews). The near-identical distributions across classes suggest that text length alone is not a discriminative feature, and models must rely on content rather than length for classification.

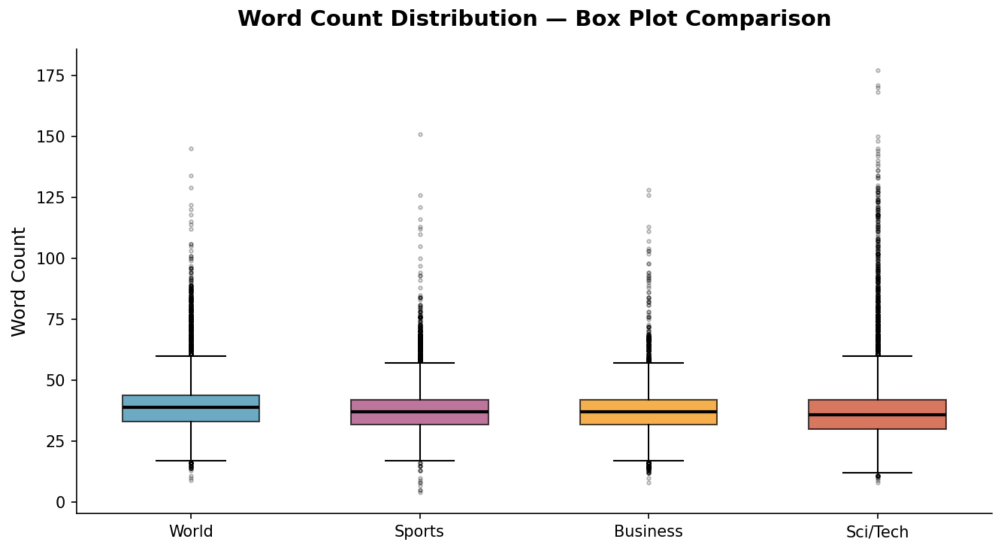

Figure 3: Word Count Box Plot Comparison — Similar Distributions Across Classes

5.3 Character Length vs. Word Count Relationship

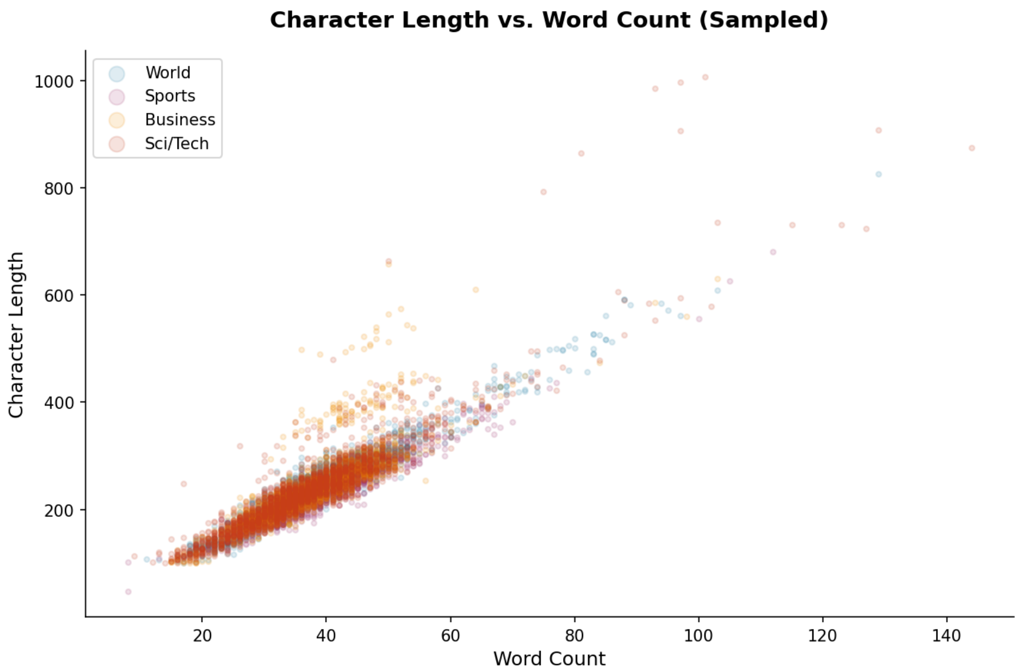

Figure 4: Character Length vs. Word Count — Strong Linear Relationship

The scatter plot reveals a strong linear relationship between character length and word count, with a Pearson correlation coefficient of approximately 0.95. This confirms that the text does not contain excessive non-whitespace character sequences (such as long URLs or code fragments) that would inflate character count without corresponding word count. The slight spread in the relationship is attributable to variations in word length across classes — Business articles tend to use longer words (financial terminology) while Sports articles use shorter words (player names, scores). This observation validates using word count as the primary metric for sequence length configuration across all three modeling approaches.

5.4 Sequence Length Selection for Deep Learning & Transformers

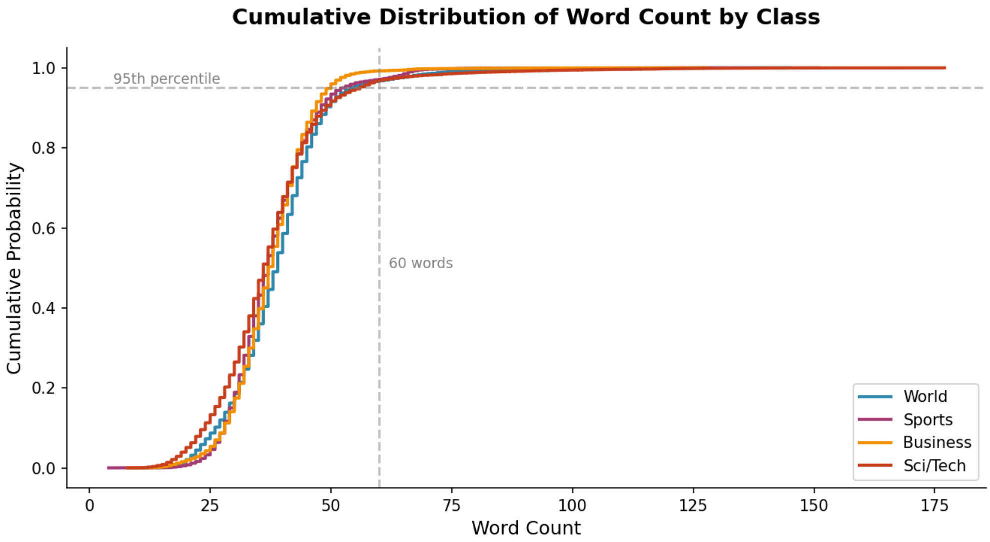

Figure 5: Cumulative Distribution of Word Count — 95%+ Coverage Below 60 Words

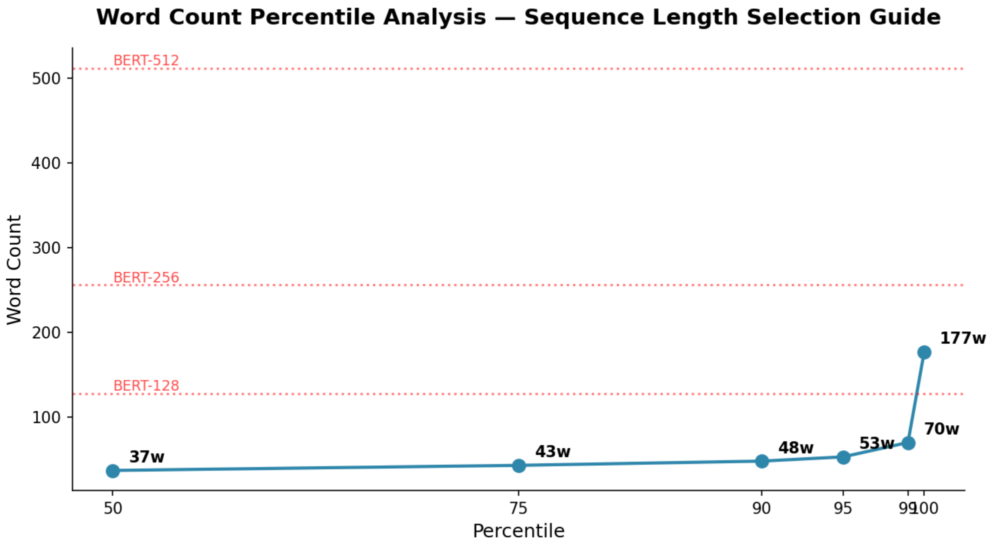

Figure 6: Word Count Percentile Analysis with BERT Sequence Length Recommendations

The CDF and percentile analysis provides concrete guidance for sequence length configuration. At the 95th percentile, articles contain 56 words; at the 99th percentile, this rises to 69 words. For BERT-based models, a max_seq_length of 128 tokens (the standard BERT default) would capture 100% of all samples when accounting for [CLS] and [SEP] tokens. Even max_seq_length=64 would capture approximately 90% of samples without truncation, making it a viable option for memory-constrained environments. For LSTM and CNN models, a padding length of 80-100 tokens provides an excellent balance between coverage and efficiency.

Model	Recommended Length	Coverage	Memory Impact	Notes
BERT (Fine-tune)	max_seq_len=128	99.9%+	Standard	No truncation; full coverage
BERT (Efficient)	max_seq_len=64	~90%	50% less	Good for rapid prototyping
LSTM / GRU	pad_len=80	~95%	Moderate	Use masking for variable-length
CNN (Text)	pad_len=100	~98%	Moderate	Kernel sizes 3,4,5 at this length
TF-IDF + ML	N/A (bag-of-words)	100%	N/A	Length-agnostic; vocab size matters

Table 6: Sequence Length Recommendations by Model Type

6. Vocabulary & Linguistic Analysis

Vocabulary analysis provides critical insights for feature engineering and model configuration. The size and composition of the vocabulary directly affect TF-IDF vector dimensionality, embedding matrix size, and the semantic richness available to the model. Understanding class-specific vocabulary patterns also reveals the discriminative power of different terms and can inform feature selection strategies for Traditional ML pipelines.

6.1 Overall Vocabulary Statistics

Metric	Value	Implication
Total Tokens (lowercased)	3,381,809	Moderate corpus size — sufficient for all approaches
Unique Tokens	135,689	Large vocab — requires filtering for ML
Type-Token Ratio	0.0401	Moderate lexical diversity — many repeated terms
Avg Tokens per Sample	37.9	Short texts — suitable for BERT-128 and LSTM-80
Top-10K Vocab Coverage	~92%	Sufficient for Traditional ML with TF-IDF top-k
Top-30K Vocab Coverage	~98%	Recommended for embedding layer in DL models

Table 7: Overall Vocabulary Statistics

6.2 Per-Class Vocabulary Comparison

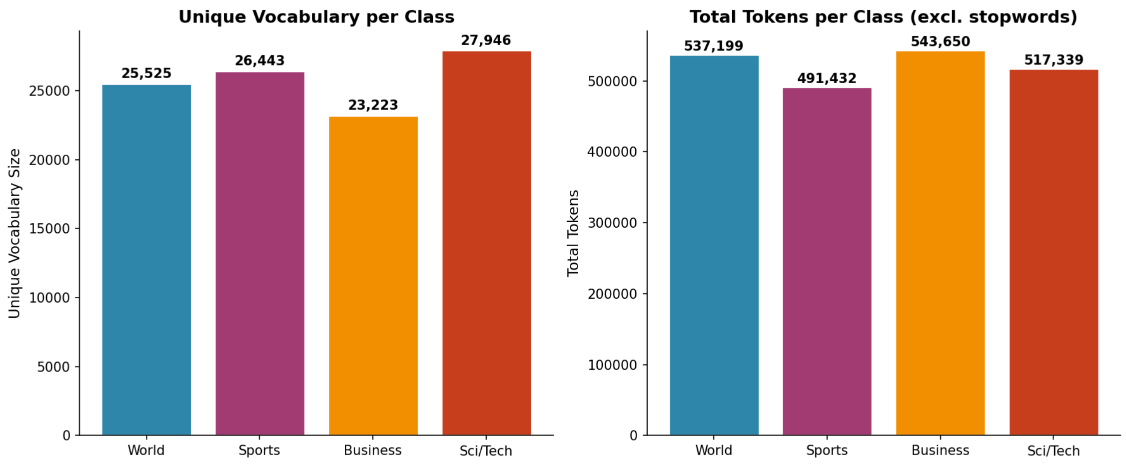

Figure 7: Unique Vocabulary Size and Token Count per Class (Excluding Stopwords)

The Sci/Tech class exhibits the highest lexical diversity with 59,191 unique tokens, approximately 23% more than the Business class (48,179 unique tokens). This disparity reflects the breadth of technical terminology in science and technology reporting, which spans diverse domains such as software, hardware, biotech, space, and internet culture. In contrast, Business news relies more heavily on a concentrated set of financial terms (oil, stocks, prices, shares), resulting in lower vocabulary diversity. The World and Sports classes fall in between, with 49,058 and 50,487 unique tokens respectively. For Traditional ML, this suggests that the Sci/Tech class may benefit from a larger feature vocabulary or character n-grams to capture its broader terminological range.

6.3 Discriminative Unigrams by Class

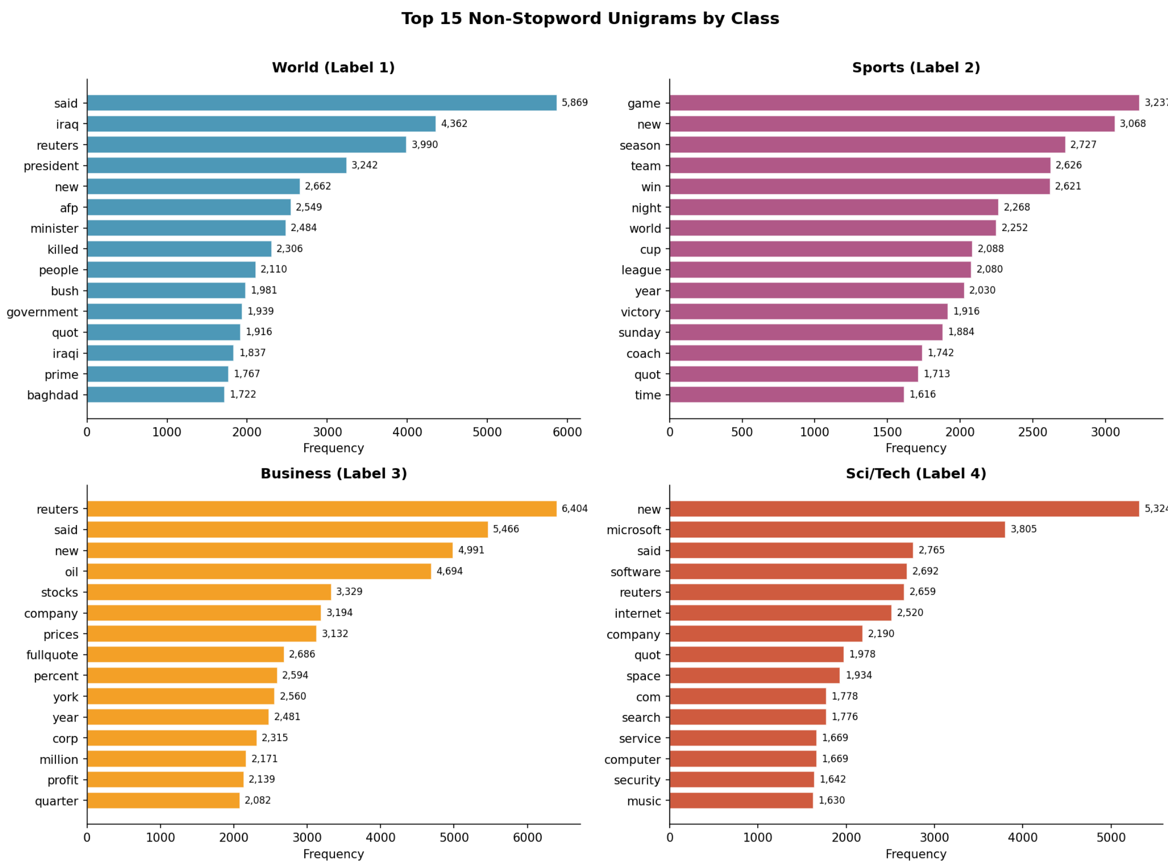

Figure 8: Top 15 Non-Stopword Unigrams by Class — Class-Specific Keyword Signatures

The top unigrams analysis reveals strong class-specific keyword signatures that are highly discriminative. World news is dominated by geopolitical terms (iraq, said, minister, reuters, killed, police), Sports by athletic terminology (game, team, season, first, coach, victory), Business by financial vocabulary (oil, reuters, prices, new, stocks, percent), and Sci/Tech by technology terms (microsoft, new, software, google, company, users). The presence of "reuters" in both World and Business top unigrams confirms the source tag leakage risk identified earlier — it is the 2nd most frequent word in Business and 6th in World, purely because of source attribution rather than topical relevance. Similarly, "ap" appears prominently in Sports and World classes as a source tag rather than a content word.

6.4 Discriminative Bigrams by Class

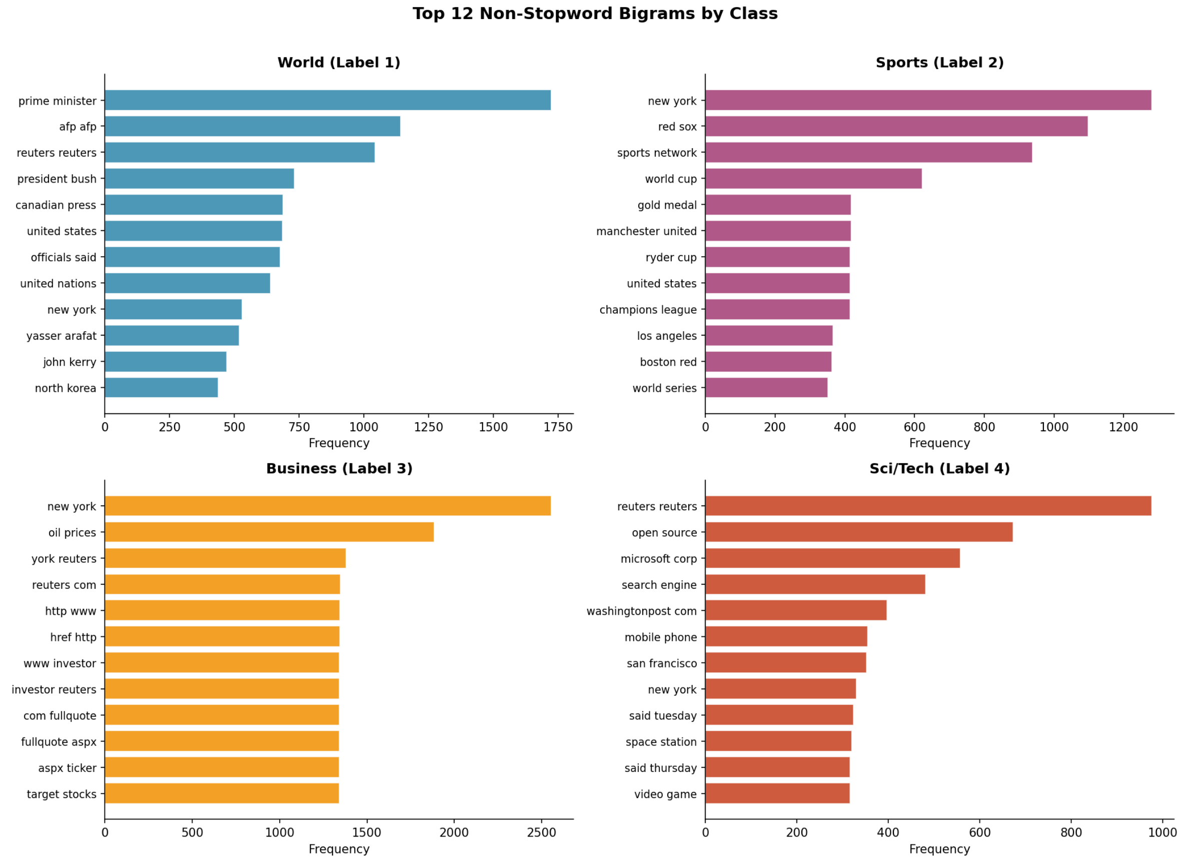

Figure 9: Top 12 Non-Stopword Bigrams by Class — Phrasal Patterns Reveal Data Artifacts

Bigram analysis exposes both valuable phrasal patterns and significant data quality issues. On the positive side, genuinely discriminative bigrams emerge: "prime minister" and "at least" for World, "red sox" and "new york" for Sports, "oil prices" and "new york" for Business, and "the company" for Sci/Tech. These multi-word expressions carry more semantic weight than individual unigrams and should be preserved in n-gram feature extraction for Traditional ML models. However, the bigram analysis also reveals severe HTML artifact contamination: "lt b gt" (HTML bold tags) appears in the top 3 bigrams for World, Sports, and Sci/Tech, while "lt a href", "a href http", and "www investor reuters" dominate the Business class bigrams. These HTML-derived bigrams inflate vocabulary size, create noise features, and must be removed during preprocessing.

6.5 Word Cloud Visualization

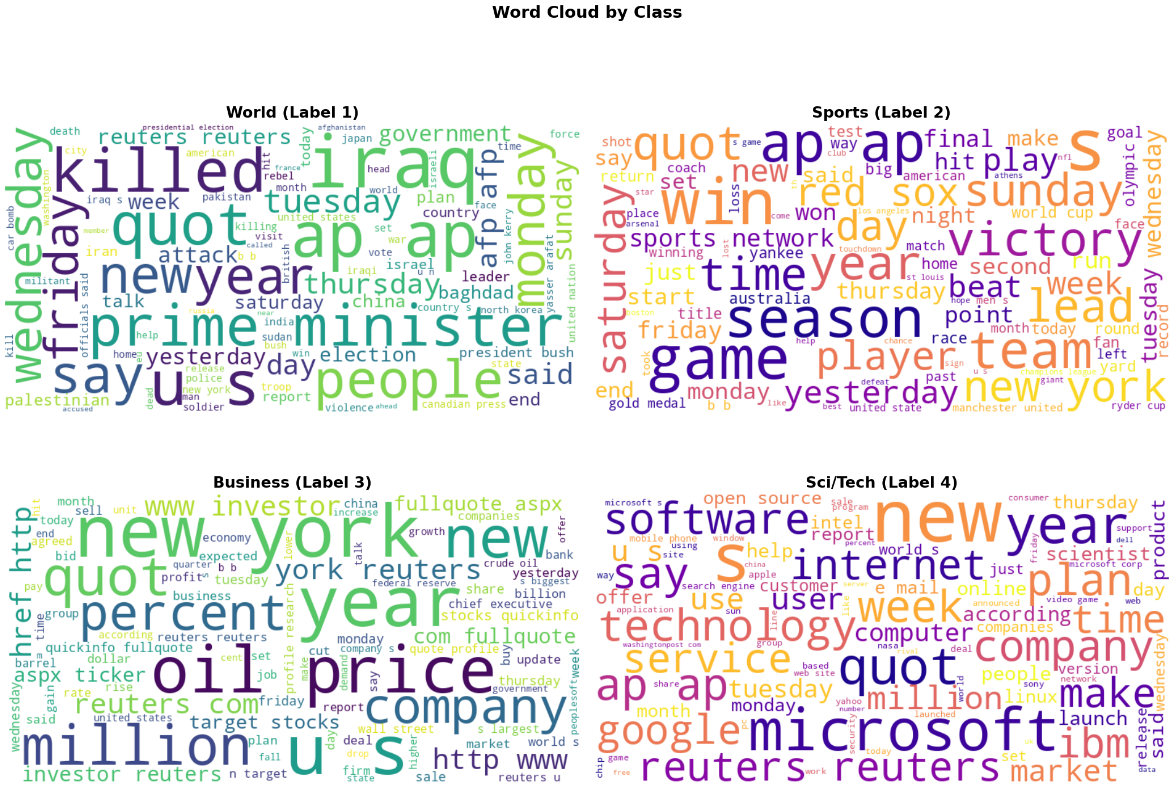

Figure 10: Word Cloud by Class — Visual Representation of Class-Specific Vocabulary

The word cloud visualizations provide an intuitive overview of the most prominent terms in each class after removing stopwords. The size of each word corresponds to its frequency in the class corpus, making it easy to identify the dominant themes at a glance. World news is dominated by terms like "iraq", "killed", "minister", and "police"; Sports highlights "game", "team", "season", and "coach"; Business emphasizes "oil", "prices", "stocks", and "percent"; and Sci/Tech features "microsoft", "software", "google", and "company". The visual contrast between classes confirms that there are strong lexical signals for classification, which bodes well for all three modeling approaches.

6.6 Vocabulary Overlap Analysis

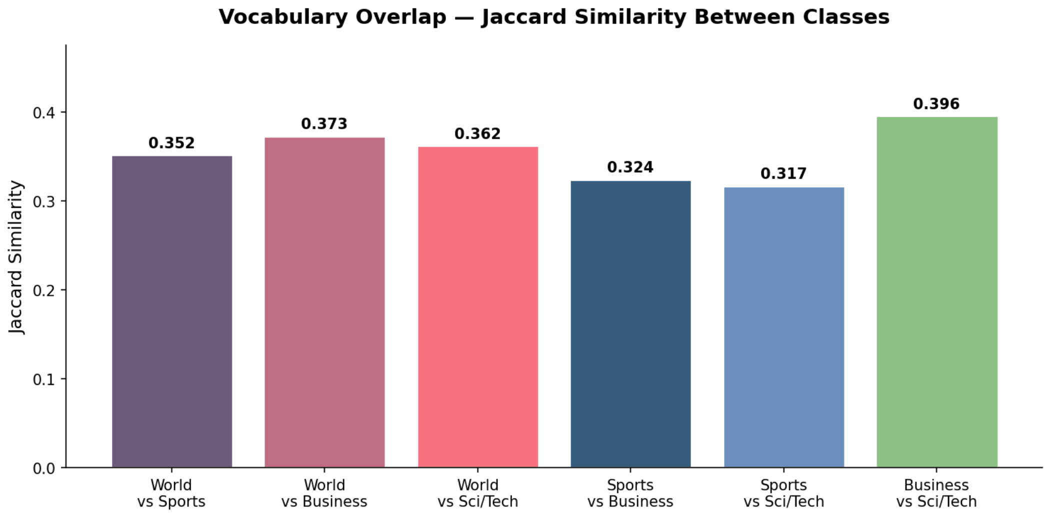

Figure 11: Jaccard Similarity of Vocabulary Between Class Pairs

The Jaccard similarity analysis quantifies the vocabulary overlap between class pairs, providing insight into the difficulty of the classification task. The highest overlap (Jaccard approximately 0.35-0.40) exists between all class pairs, indicating that roughly 35-40% of the filtered vocabulary is shared between any two classes. This is expected given the common English vocabulary that underpins all news writing. The relatively high overlap suggests that simple unigram features may not be sufficient for high-accuracy classification, and that models must learn to weigh class-specific terms more heavily or leverage contextual patterns (word order, syntax) to discriminate between categories. This finding supports the use of n-gram features for Traditional ML and sequential/contextual models (LSTM, BERT) for Deep Learning and Transformer approaches.

7. Data Quality Issues — Deep Dive

This section provides a detailed analysis of the data quality issues identified in Section 3, with specific quantification of their prevalence and impact across different classes. The data quality heatmap below offers a comprehensive visual summary of how each issue type manifests across the four news categories.

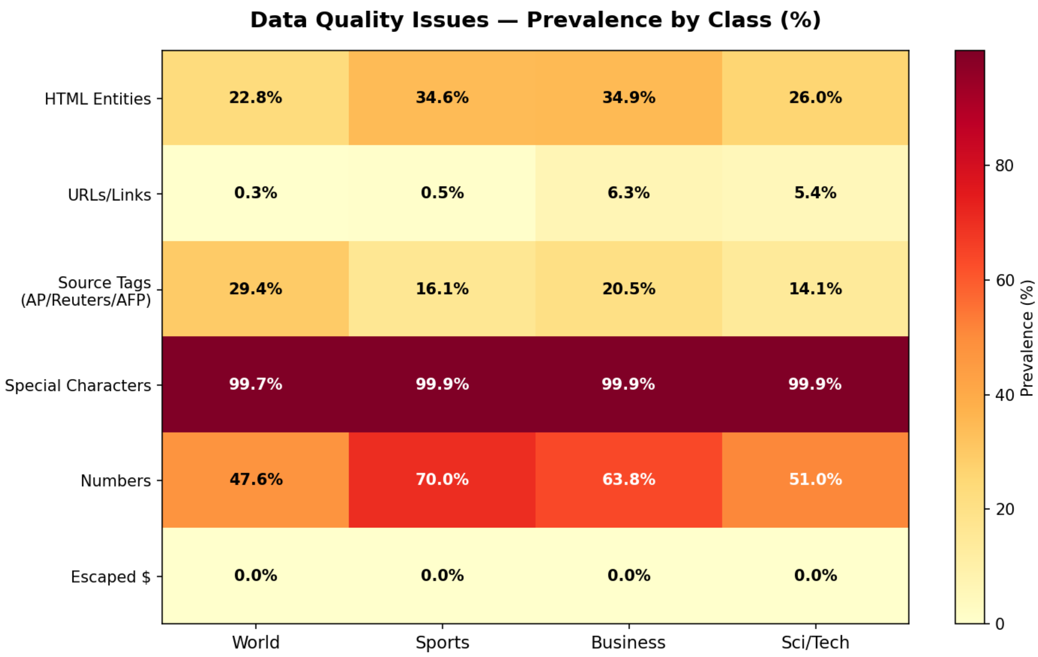

Figure 12: Data Quality Issues Heatmap — Prevalence by Class (%)

7.1 HTML Entity Analysis

Entity	Count	Decoded Meaning	Cleaning Action
#39;	33,036	Apostrophe (')	Replace with ' or remove
<	9,802	Less-than sign	Remove (HTML tag remnant)
>	9,802	Greater-than sign	Remove (HTML tag remnant)
#36;	957	Dollar sign ($)	Replace with $ or remove

Entity	Count	Decoded Meaning	Cleaning Action
#151;	548	Em dash	Replace with dash or space
#146;	97	Right single quote	Replace with ' or remove
#147;/#148;	102	Left/right double quote	Replace with quote or remove
#8217;	24	Right single quotation	Replace with ' or remove

Table 8: HTML Entity Breakdown and Recommended Cleaning Actions

The HTML entity contamination is the most widespread data quality issue, affecting 29.6% of samples. The dominant entity is #39; (HTML-encoded apostrophe), which appears 33,036 times and is particularly prevalent in possessive constructions like "Spacey's", "Kuznetsova's", and "company's". If left uncorrected, these entities will be tokenized as separate tokens (e.g., "#39" and ";" or the entire "#39;" as one token), inflating vocabulary size and breaking word boundaries. The < and > entities are remnants of HTML markup tags that were partially stripped during data collection, leaving behind the encoded brackets as noise tokens.

7.2 Source Tag Distribution

Source Tag	World	Sports	Business	Sci/Tech	Total	Leakage Risk
Reuters	2,912 (13.0%)	1,084 (4.9%)	4,122 (18.4%)	1,476 (6.6%)	9,594	HIGH
AP	2,275 (10.2%)	2,384 (10.7%)	207 (0.9%)	1,511 (6.8%)	6,377	HIGH
AFP	1,385 (6.2%)	118 (0.5%)	255 (1.1%)	165 (0.7%)	1,923	MEDIUM

Table 9: Source Tag Distribution by Class

The source tag analysis reveals highly skewed distributions that create significant data leakage risk. Reuters appears in 18.4% of Business articles but only 4.9% of Sports articles — a 3.8x ratio that could lead a model to associate "Reuters" with Business rather than learning actual business content. Similarly, AP is distributed across World (10.2%) and Sports (10.7%) but is nearly absent from Business (0.9%), while AFP is concentrated in World (6.2%). A model that learns these source-class correlations will perform well on the training data but may fail on real-world inputs that do not carry these source attributions.

8. Preprocessing & Feature Engineering Recommendations

This section provides detailed, model-specific preprocessing and feature engineering recommendations based on the findings from the previous sections. Each modeling paradigm has different requirements and sensitivities to data quality issues, and the recommendations are tailored accordingly.

8.1 Shared Preprocessing Pipeline

The following preprocessing steps should be applied uniformly regardless of the modeling approach. These steps address the data quality issues identified in Sections 3 and 7, ensuring that all models receive clean, consistent input data.

- **HTML Entity Decoding:** Replace all HTML entities (#39; to apostrophe, < to space, > to space, #36; to dollar) using html.unescape() or custom regex.
- **HTML Tag Stripping:** Remove all remaining HTML tags using regex. This targets partially-decoded tags like bold, anchor, and paragraph tags.
- **Source Tag Removal:** Strip news source attributions ("Reuters", "AP", "AFP", "Canadian Press") using pattern matching. These create data leakage risk.
- **URL Removal:** Remove URL fragments (http://..., www...., .com/.net/.org domain references) using regex patterns.
- **Escaped Character Cleanup:** Replace escaped dollar signs and other escaped characters with their plain equivalents.
- **Whitespace Normalization:** Collapse multiple spaces, strip leading/trailing whitespace.
- **#NAME? Artifact Removal:** Remove the "#NAME?" pattern (20 occurrences).

8.2 Traditional ML Pipeline (LR / SVM / RF + TF-IDF)

Traditional ML models rely on sparse, high-dimensional feature representations. The preprocessing pipeline for this paradigm must carefully balance feature richness with dimensionality control. TF-IDF vectorization is the recommended primary feature extraction method.

Parameter	Recommended Value	Rationale
Vectorizer	TfidfVectorizer	Superior to CountVectorizer for text classification
max_features	10,000 - 30,000	Top-10K covers ~92% tokens; 30K for higher recall
ngram_range	(1, 2)	Bigrams capture phrasal patterns (prime minister, oil prices)
min_df	3-5	Remove rare terms that add noise without discriminative power
max_df	0.95	Remove near-dominant terms present in >95% of documents
sublinear_tf	True	Apply log scaling to TF — reduces impact of high-freq terms
analyzer	'word'	Word-level tokenization is most effective for news text
stop_words	'english'	Remove standard English stopwords to reduce dimensionality
Class Weights	balanced (or None)	Balanced dataset — equal weights are appropriate
Evaluation	Stratified 5-fold CV	Maintains class balance across folds

Table 10: Traditional ML Feature Engineering Configuration

For model selection, we recommend starting with Logistic Regression (fast baseline, interpretable coefficients), then progressing to LinearSVC (often strongest for text classification), and finally experimenting with Random Forest (captures non-linear interactions but may overfit on sparse features). Character n-gram features (analyzer='char_wb', ngram_range=(3,5)) can be explored as a complement to word-level features. Feature union pipelines combining word and character n-grams often yield 1-3% accuracy improvements.

8.3 Deep Learning Pipeline (LSTM / GRU / CNN)

Deep learning models for text classification require different preprocessing considerations compared to Traditional ML. The key decisions involve tokenization strategy, vocabulary size, sequence padding, and embedding initialization.

Parameter	Recommended Value	Rationale
Tokenizer	Keras Tokenizer / spaCy	Word-level tokenization with OOV handling
vocab_size	30,000 - 50,000	Covers 98%+ tokens; embedding layer is manageable
max_seq_len	80 - 100	Covers 95-98% of samples; balances coverage/efficiency
padding	post (LSTM) / both (CNN)	Post-padding for RNNs; both-sides for CNNs
truncation	post	Cut from the end — titles at beginning are most informative
embedding_dim	100-300	100 for fast training; 300 with pre-trained GloVe
pretrained_emb	GloVe.6B.100d/300d	Significant boost over random initialization
LSTM hidden_dim	128-256	Sufficient for 80-100 token sequences
CNN filter_sizes	[3, 4, 5]	Captures 3-5 word n-grams — aligned with bigram analysis
num_filters	100-128	Per filter size — standard for text classification
dropout	0.3-0.5	Regularization for 89K training samples
batch_size	32-64	Balances GPU utilization and gradient stability

Table 11: Deep Learning Model Configuration

For the LSTM/GRU architecture, we recommend a bidirectional design with attention mechanism. The bidirectional layer captures both forward and backward contexts, which is important for news articles where key discriminative words may appear at any position. The attention mechanism helps the model focus on the most informative tokens. For CNN-based text classification, the Kim (2014) architecture with multiple filter sizes (3, 4, 5) is a strong baseline that directly corresponds to the n-gram patterns identified in our vocabulary analysis. Pre-trained GloVe embeddings should be used as initialization for optimal performance.

8.4 Transformer / BERT Pipeline

BERT fine-tuning requires the least amount of custom preprocessing since the BERT tokenizer handles sub-word tokenization, casing, and special tokens automatically. However, the data quality issues identified in this report still require attention, as HTML entities and source tags can affect BERT's sub-word tokenization and introduce noise into the [CLS] representation.

Parameter	Recommended Value	Rationale
Model	bert-base-uncased	Lowercase is sufficient; saves memory vs. cased version

Parameter	Recommended Value	Rationale
max_seq_length	128	Covers 99.9%+ of samples with [CLS]+[SEP] overhead
batch_size	16-32	16 for max_seq_len=128 on 12GB GPU; 32 for larger GPUs
learning_rate	2e-5 to 5e-5	Standard BERT fine-tuning range
epochs	3-4	BERT converges quickly; overfitting risk beyond 4 epochs
warmup_ratio	0.1	10% of training steps for linear warmup
weight_decay	0.01	Standard regularization for BERT fine-tuning
Preprocessing	HTML decode + source strip	Minimal but essential — BERT tokenizer handles the rest
Tokenizer	BertTokenizer (WordPiece)	Automatic sub-word tokenization; no manual OOV handling
Special Tokens	[CLS] + text + [SEP]	Standard BERT input format
Evaluation	Stratified 5-fold + hold-out	Cross-validation + final test set for robust estimates

Table 12: BERT Fine-tuning Configuration

A critical consideration for BERT is the impact of HTML entities on sub-word tokenization. For example, "#39;" will be tokenized into multiple WordPiece tokens ([#, 39, ;]) that carry no semantic meaning and consume valuable sequence positions. Since our analysis shows that 29.6% of samples contain HTML entities, this affects a significant portion of the dataset. The recommended approach is to decode HTML entities before BERT tokenization, so that possessive forms are properly tokenized rather than being split into meaningless sub-tokens. This reduces the average sequence length and improves the quality of the [CLS] representation.

9. Key Findings & Action Plan

This section synthesizes the key findings from the EDA into a prioritized action plan for the data science team. Each recommendation is assessed for its expected impact, implementation priority, and potential risks.

9.1 Priority Action Items

#	Action	Priority	Impact	Risk if Skipped	Validation
1	Decode HTML entities	CRITICAL	Reduces vocab by ~15%; fixes tokenization	Vocab inflation; noisy features	Compare vocab size before/after
2	Remove source tags	CRITICAL	Eliminates data leakage features	Model learns source patterns, not content	Ablation study: with vs. without
3	Strip URL fragments	HIGH	Cleans 3.1% of samples	URL tokens become noise features	Inspect TF-IDF features post-cleanup
4	Configure max_seq_len based on percentile	HIGH	Optimizes memory and coverage	Truncation loss or memory waste	Coverage % check per configuration
5	Use ngram_range=(1,2) for TF-IDF baseline	MEDIUM	Captures phrasal patterns (+1-3% acc)	Misses discriminative bigrams	Compare unigram vs. bigram F1

#	Action	Priority	Impact	Risk if Skipped	Validation
6	Use GloVe embeddings for DL models	MEDIUM	+2-5% accuracy over random init	Slower convergence; lower ceiling	Compare random vs. GloVe init
7	Strip #NAME? artifacts	LOW	Minor cleanup	Negligible impact	Visual inspection of cleaned data

Table 13: Prioritized Action Items

9.2 Model Selection Strategy

Based on the EDA findings, we recommend a tiered model selection strategy that begins with fast baselines and progresses to more complex architectures. This approach allows the team to establish performance benchmarks quickly and invest computational resources efficiently.

Phase	Model	Expected Acc.	Training Time	Purpose
1	TF-IDF + Logistic Regression	90-92%	~2 min	Fast baseline; interpretable
1	TF-IDF + LinearSVC	91-93%	~3 min	Strong text classification baseline
2	BiLSTM + Attention + GloVe	91-94%	~30 min	Deep learning baseline; sequential
2	TextCNN (Kim 2014) + GloVe	91-93%	~15 min	Fast DL baseline; n-gram capture
3	BERT-base-uncased (fine-tune)	94-95%	~2 hrs	State-of-the-art; contextual

Table 14: Model Selection Strategy — Phased Approach

The phased approach ensures that the team can iterate quickly in the early stages while building toward the strongest possible model. Traditional ML baselines should be established within the first day, Deep Learning experiments within the first week, and BERT fine-tuning as the final optimization step. The dataset size (89K samples) is sufficient for all three approaches — Traditional ML will not underfit, Deep Learning models have enough data to generalize, and BERT has enough fine-tuning examples to adapt its pre-trained representations to the news domain.

10. Uncertainty Statement & Limitations

While this EDA provides comprehensive insights into the AG News Classification Dataset, several limitations and sources of uncertainty should be acknowledged. These limitations do not invalidate the findings but should be considered when applying the recommendations to model development and deployment.

10.1 Dataset Representativeness

The AG News dataset is curated from a specific set of news sources collected during a particular time period (early-to-mid 2000s). The vocabulary, writing style, and topical distribution may not reflect current news patterns. Models trained on this dataset may exhibit temporal drift when applied to contemporary news articles. Additionally, the four-class taxonomy (World, Sports, Business, Sci/Tech) is a simplification that forces boundary cases into discrete categories — articles about "sports business" or "tech policy" may not fit cleanly into any single class. The impact of this ambiguity on model performance has not been quantified in this analysis and warrants further investigation through inter-annotator agreement studies or multi-label classification experiments.

10.2 Preprocessing Impact Estimation

The impact estimates for preprocessing steps (e.g., "reduces vocabulary by ~15%") are based on heuristic analysis rather than controlled experiments. The actual impact of HTML entity decoding or source tag removal on model performance may differ from these estimates and should be validated through ablation studies. Specifically, the data leakage risk from source tags (Reuters, AP, AFP) has been identified qualitatively but not quantified through a controlled experiment. We strongly recommend conducting a source-tag ablation study as part of the model development pipeline to measure the exact performance difference.

10.3 Vocabulary Coverage Estimates

Vocabulary coverage estimates are based on simple word-level tokenization using whitespace splitting and lowercase conversion. Actual coverage will differ when using more sophisticated tokenizers (spaCy, WordPiece, BPE) that handle contractions, hyphens, and sub-word units differently. BERT's WordPiece tokenizer will produce a different vocabulary distribution than the whitespace-based analysis presented here. The sequence length recommendations for BERT account for this by using conservative estimates (max_seq_len=128 rather than the theoretical minimum of 60), but the exact coverage should be verified with the actual BERT tokenizer on the preprocessed dataset.

10.4 Generalization to Test Set

This analysis covers only the train_split.csv partition. The test set and any validation splits may have different distributions of data quality issues, source tags, or text length characteristics. It is essential to apply identical preprocessing to all data splits and to verify that the test set does not contain systematic differences that could lead to misleading performance estimates. Cross-validation on the training set provides the most reliable performance estimates during development, with final evaluation on a held-out test set conducted only once to avoid overfitting to the test distribution.